CS 4039 Software for Mobile Devices

Assignment # 2 — Kinnect Backend with Firebase

Due Date: As mentioned on GCR

Instructions:

This is a **two-person group assignment**. Only one group member needs to submit the assignment on behalf of the team. Submissions that do not comply with the specifications given in this document will not be marked, and a zero mark will be assigned.

- Export and upload your complete project as a ZIP file using **Android Studio → File → Export → Export to ZIP File**.
- The ZIP / folder name must follow the format **rollNumber1_rollNumber2_Assignment02** (e.g., **i220001_i220002_Assignment02**).
- The project name must be **rollNumber1_rollNumber2** (e.g., **i220001_i220002**), and the package name must be **com.groupname.rollNumber1_rollNumber2** (e.g., **com.teamalpha.i220001_i220002**).
- Each team must use a **private GitHub repository** so others cannot view or copy your code.
- Commits from **both** team members are mandatory; the commit history must clearly reflect each person's contribution.
- Do **not** commit your **google-services.json** or Agora App Certificate to a public location.

Introduction:

In Assignment 1 you built the complete front end of **Kinnect** — 23 pixel-perfect screens with working navigation. In this assignment you will bring Kinnect to life by connecting it to **Firebase** as its backend. Kinnect uses **Firebase Authentication** for accounts, **Firebase Realtime Database** for users, posts, reactions, comments, stories, friend requests, chats, call logs and presence, and **Firebase Cloud Messaging (FCM)** for push notifications. Real-time voice and video calls are powered by the **Agora SDK**.

Every screen from Assignment 1 must now display **live data** instead of hard-coded content. The UI must continue to match Kinnect_UI.pdf exactly. Where a feature needs a screen that is not in the design (see the table at the end), you must create it in the same Kinnect design language (primary teal **#0B5F63**, same typography, spacing and corner radii).

Tasks:

GitHub Version Control (10 Marks)

- Push your project to the private repository and commit regularly as features are completed.
- Each implemented feature must have its own commit **with a meaningful message.**
- **Commits from both group members are mandatory.** Late commits (after the deadline) will not count.

Splash Screen (5 Marks)

- Splash Screen must remain visible for **5 seconds** and then navigate automatically.
- If a user is already logged in, Splash must go **directly to Home;** otherwise it opens Log in.

User Authentication (5 Marks)

- Implement sign up, log in and log out using **Firebase Authentication** (email & password).
- Sign up (screen 03) must save the user's info to the Realtime Database. Password strength must update as the user types.
- **First-time users** must be directed to the profile setup step (Edit profile, screen 16) to choose a unique username and bio before reaching Home.
- The **Recent login** card on Log in must show the last account that signed in on the device; tapping it pre-fills the email.
- Log out (Menu, screen 19) must sign the user out, mark them offline, and **clear the back stack**.

Stories (10 Marks)

- Users can upload **image or video** stories from the Camera (09) or gallery, add a text label in the Story editor (10), and share to **Your story** or **Close friends**.
- Stories must **disappear automatically after 24 hours**
- Stories from friends must appear at the top of Home in the **horizontal scroll view**, with “Create story” first.
- Use **Firebase Realtime Database** to store and manage stories.

Posts, Reactions & Comments (10 Marks)

- Users can create posts with text and **one or more images** (Create post 07, Photo picker 08). The Home feed must render the multi-image grid with the “+N” overlay exactly as in screen 04.
- Posts support all **six reactions** through the reaction picker (05) — one reaction per user per post, which can be changed or removed. Reaction summary icons and counts must update in real time.
- Comments (06) must support **replies**, comment likes, and image comments. “Replying to” / Cancel must work.
- The feed must show posts from the user and their friends, newest first.

Messaging System (10 Marks)

- Chats (20) must list real conversations with the last message, time, and unread indicator.
- Media sharing: users can send **text, images and posts** in a Chat (21).
- **Edit & delete:** a user can edit or delete their own message **within 5 minutes** of sending it. After 5 minutes these options must not appear. Edited messages show "Edited".
- Message replies and the heart reaction on a message (as in screen 21) must be stored and displayed.

Voice & Video Calls (10 Marks)

- Implement real-time **audio and video calls** using the **Agora SDK**.
- The phone and video icons in a Chat (21) must start a call.

Friend System & Profiles (10 Marks)

- Users can **send, confirm and delete friend requests** from Search (13), Other profile (17), Friends (14) and Notifications (18).
- The Add friend button must reflect the current state: *Add friend → Requested → Friends*.
- Friends (14) must show the real pending-request count and "Your friends" list. Profiles must show the real friend count and, on another user's profile, the **mutual friends** count.
- Users can update their **profile picture, cover photo**, name, username and bio from Edit profile (16).

Push Notifications (10 Marks)

- Implement push notifications using **Firebase Cloud Messaging (FCM)**.
- Notify users about: **new messages.**

Search & Filters (5 Marks)

- Search (13) must find users by **username or name** as the user types, using the Realtime Database.

Online / Offline Status (5 Marks)

- Display whether users are online or offline **in real time** (green dot in Chats and “Active now” in Chat).

Security & Privacy (5 Marks)

- A user must receive a notification if **someone takes a screenshot of their chat**, and a system message (“Aisha took a screenshot”) must appear in the conversation.

Test Cases (5 Marks)

- Write **two Espresso test cases** for critical backend-connected workflows.

Live Demo

- You must present a **live demo** of your app on **two devices** (to show messaging, calls, presence and notifications between users).
- Both members must be present and able to explain any part of the implementation and make real-time modifications.

Your application must contain **all 23 screens** from Kinnect_UI.pdf, now connected to live data. If you believe any other screen is needed for a required feature, you must create it in the same design language.

Submission Checklist:

- Project, package and ZIP names follow the naming conventions above.
- All 23 screens match Kinnect_UI.pdf and show live Firebase data; additional screens follow the same design.
- Private GitHub repository and commits from both members.
- Two Espresso test cases are implemented.

Note:

Self-Learning Requirement: If a feature required for this assignment has not been covered in class (for example, Agora calls, FCM, or detecting screenshots), you are expected to learn and implement it independently using reliable resources.

GOOD LUCK!